



ENGINEERING OPTOGENETIC TOOL TO CONTROL CALCIUM HOMEOSTASIS AT ENDOPLASMIC RETICULUM

Mingjun Jiang
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Supervisor: Kozo Hamada



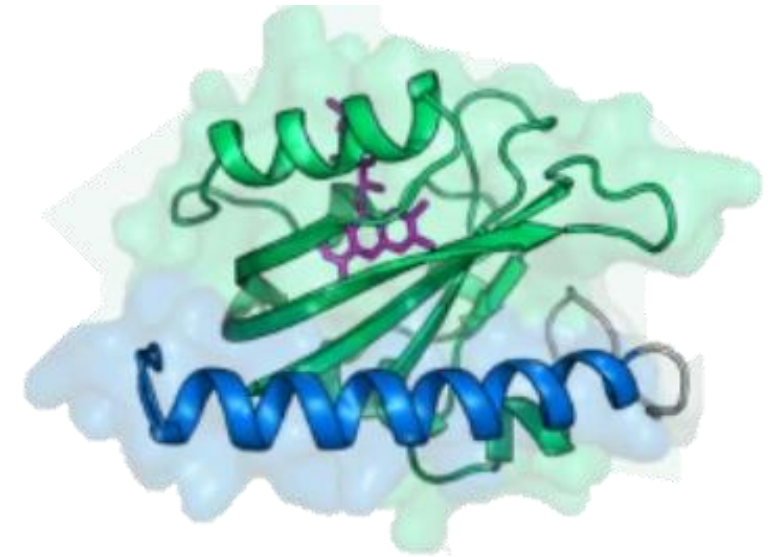
- The endoplasmic reticulum (ER) is an intracellular reservoir of calcium ions (Ca^{2+}) a critical organelle to regulate Ca^{2+} signaling
- Dysregulation of the ER Ca^{2+} causes ER stress, inducing multiple human diseases
- Sarco/endoplasmic reticulum Ca^{2+} ATPase (SERCA) is an essential membrane pump that concentrates Ca^{2+} into the ER lumen by ATP hydrolysis
- Improper function of SERCA lead to dysregulation of the ER Ca^{2+} homeostasis



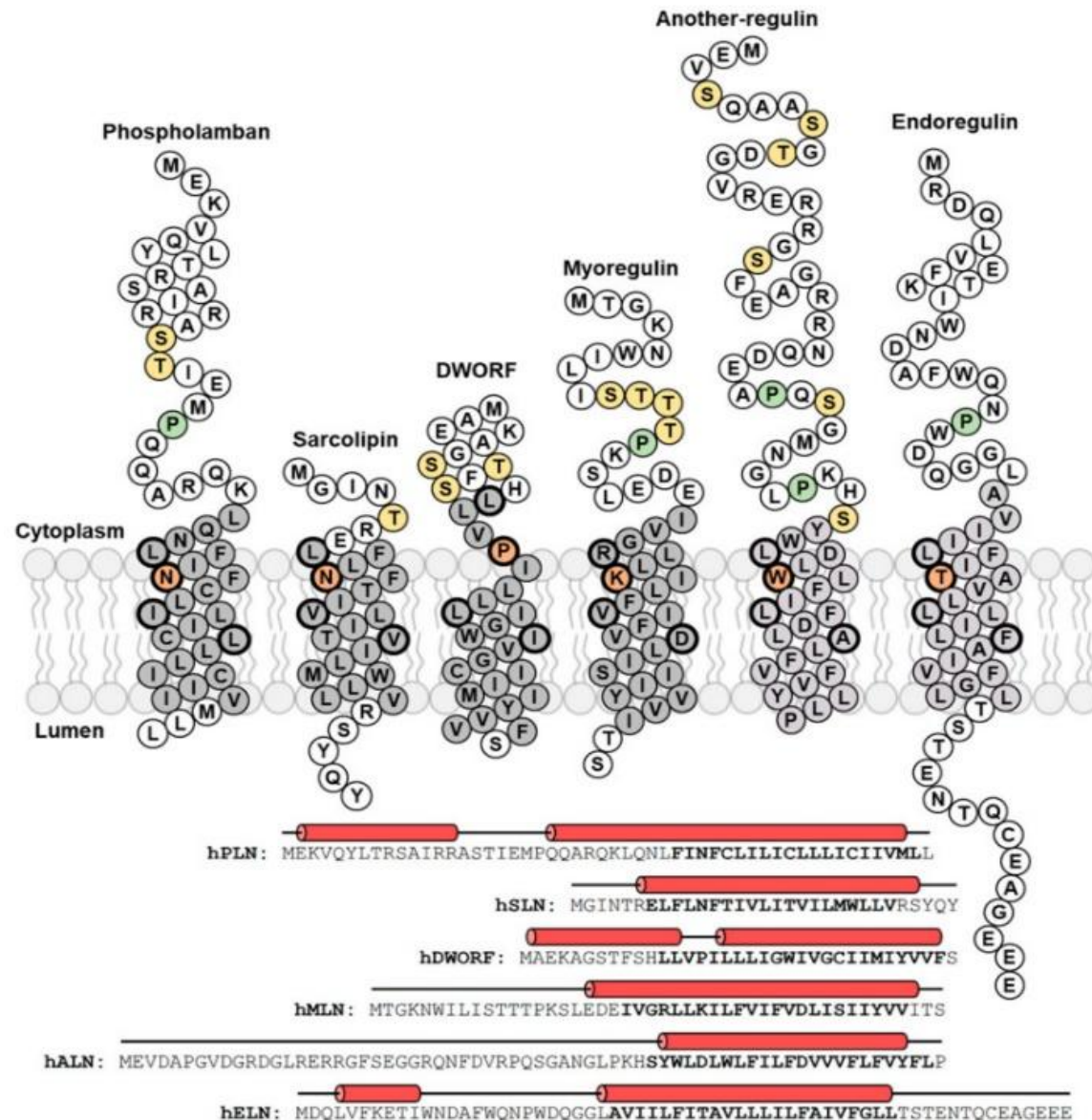
Endoplasmic Reticulum

<https://images.app.goo.gl/1Y47dRdZ9tE7yoCp6>

- LOV domains are protein photosensors conserved in various organisms that detect blue light (440-473 nm) via the endogenous flavin cofactor
- Previous research team has engineered a circularly permuted LOV2 (cpLOV2) protein and confirmed its photo-sensitivity with conformational changes
- Optogenetic tool is a promising technique to address ER stress issue

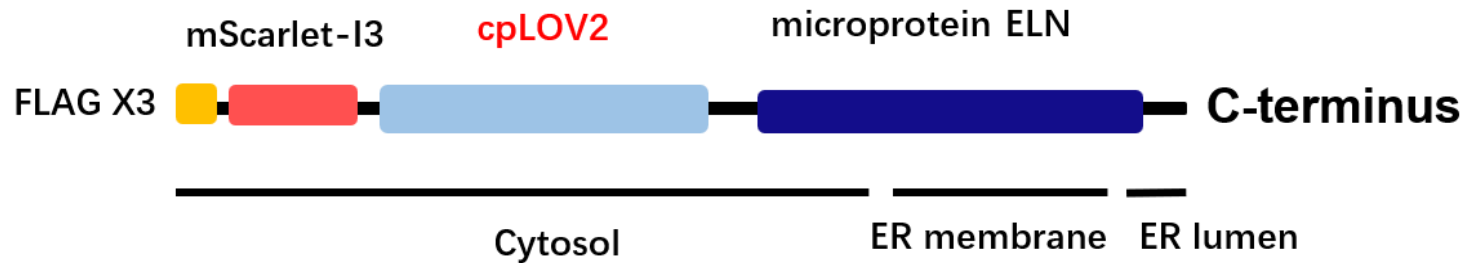


cpLOV2 3D structure

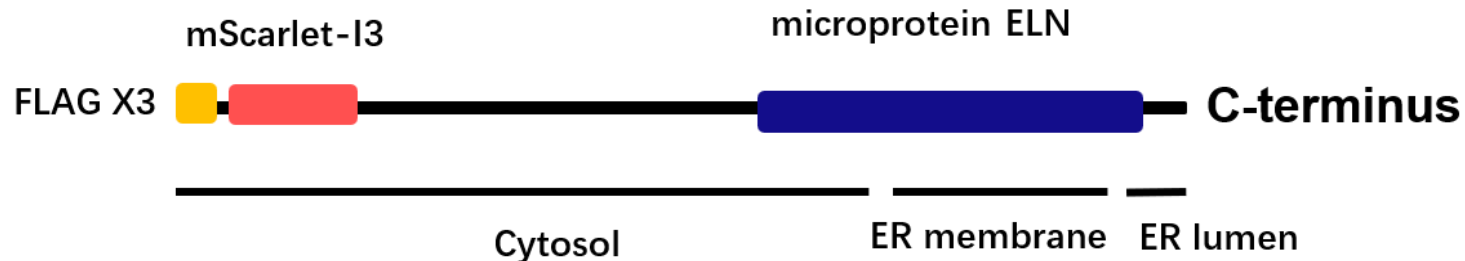


- Enduruglin (ELN) is a transmembrane microprotein, the product of non-coding RNA located on the ER membrane
- ELN interact to inhibit SERCA Ca^{2+} pump by specific binding

ELN_cpLOV2 protein



ELN_mutant protein



- Constructed an artificial optogenetic tool ELN_cpLOV2 (human ELN fusion proteins by introducing the cpLOV2 domain)
- cpLOV2 domain-deleted ELN_mutant protein also constructed as control group



Blue light-induced conformational changes of cpLOV2 domain could impact the interactions between ELN_cpLOV2 and the target SERCA pump

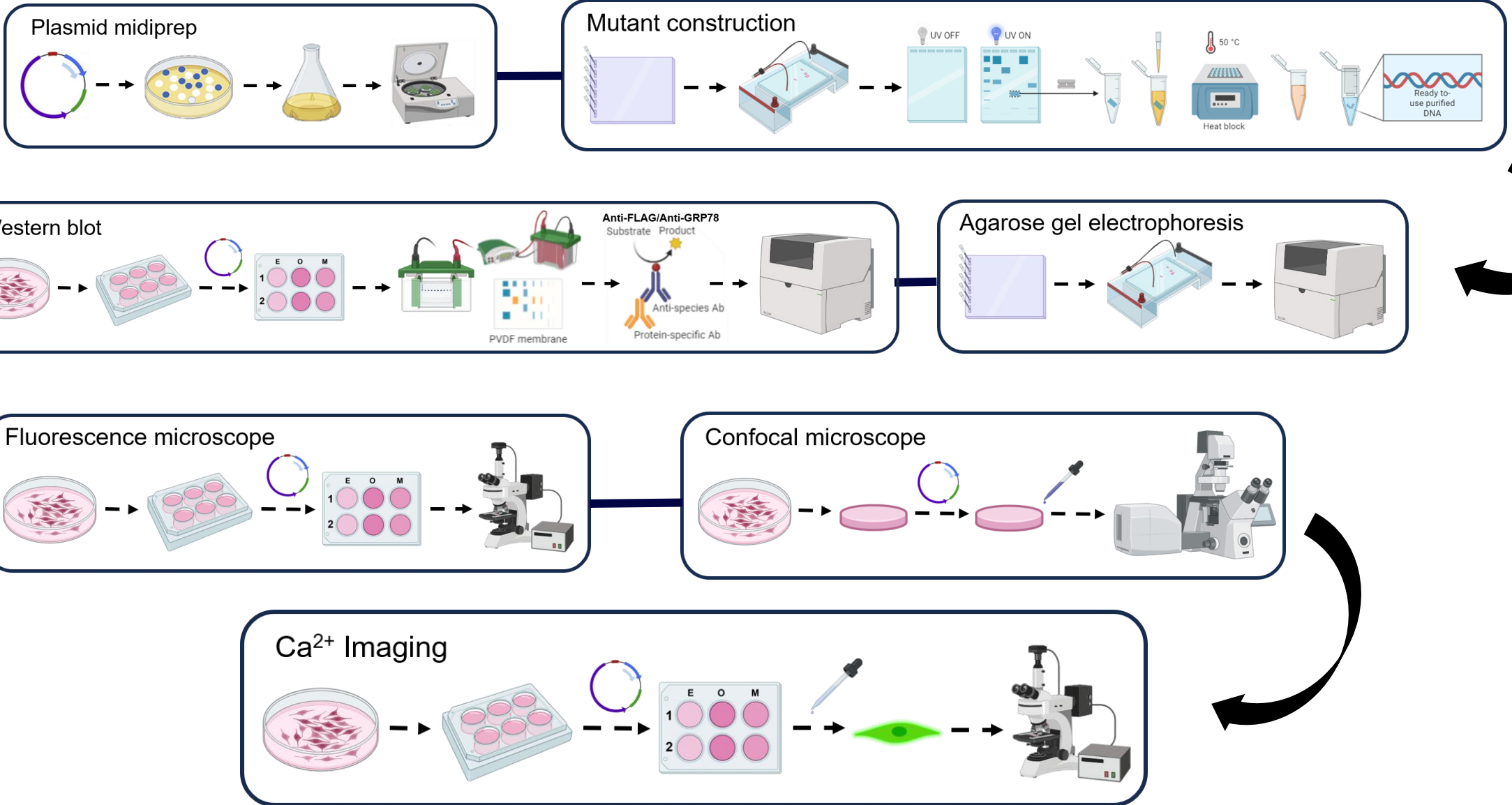


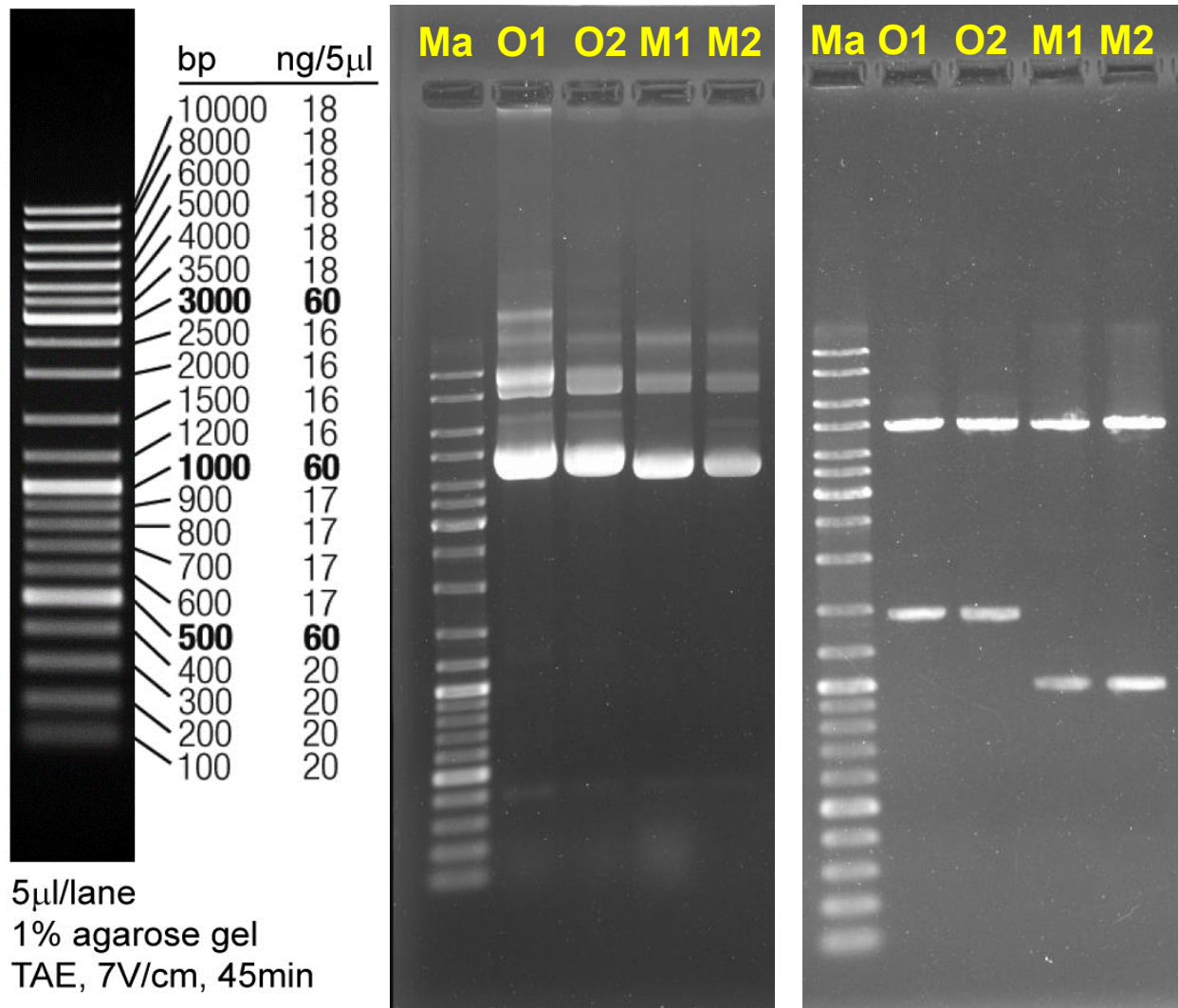
SERCA Ca²⁺ pump functions affected



Object and Project Outline

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1% agarose gel electrophoresis

Gel Electrophoresis

- Plasmid without processing and Plasmid digested by EcoRI restriction enzyme
- The plasmids of original ELN_cpLOV2 and mutants ELN_mutant were successfully constructed and generally in high amount and purity

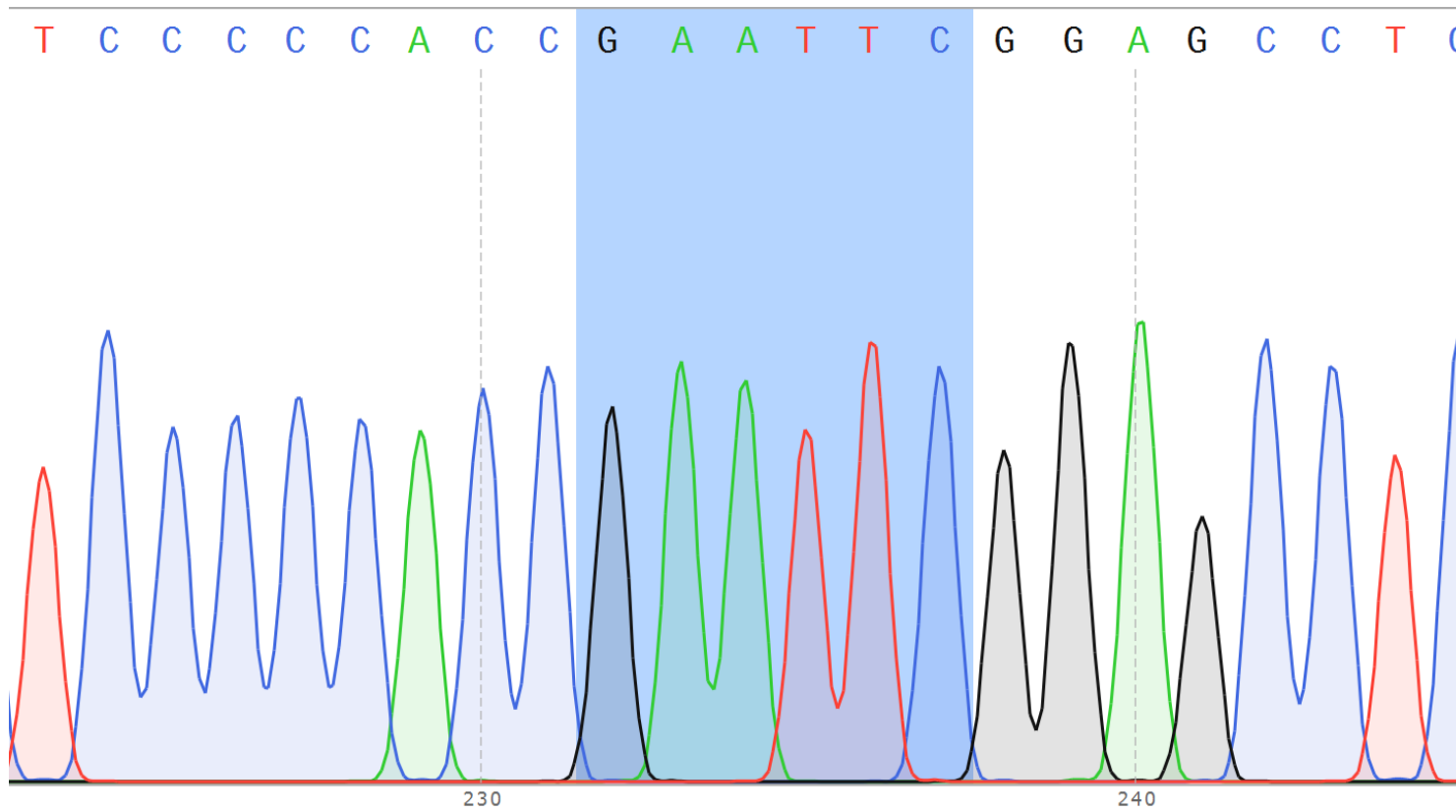


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TCCCCCACC GAATTC [REDACTED] 238> [REDACTED] GGAGCCT



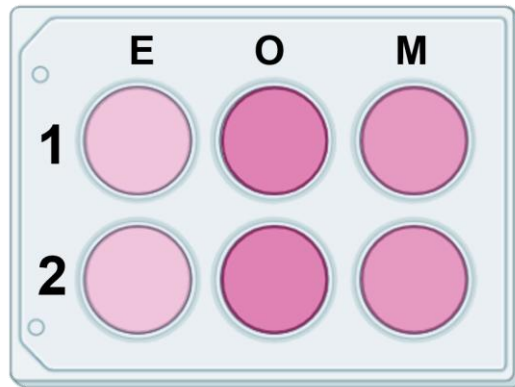
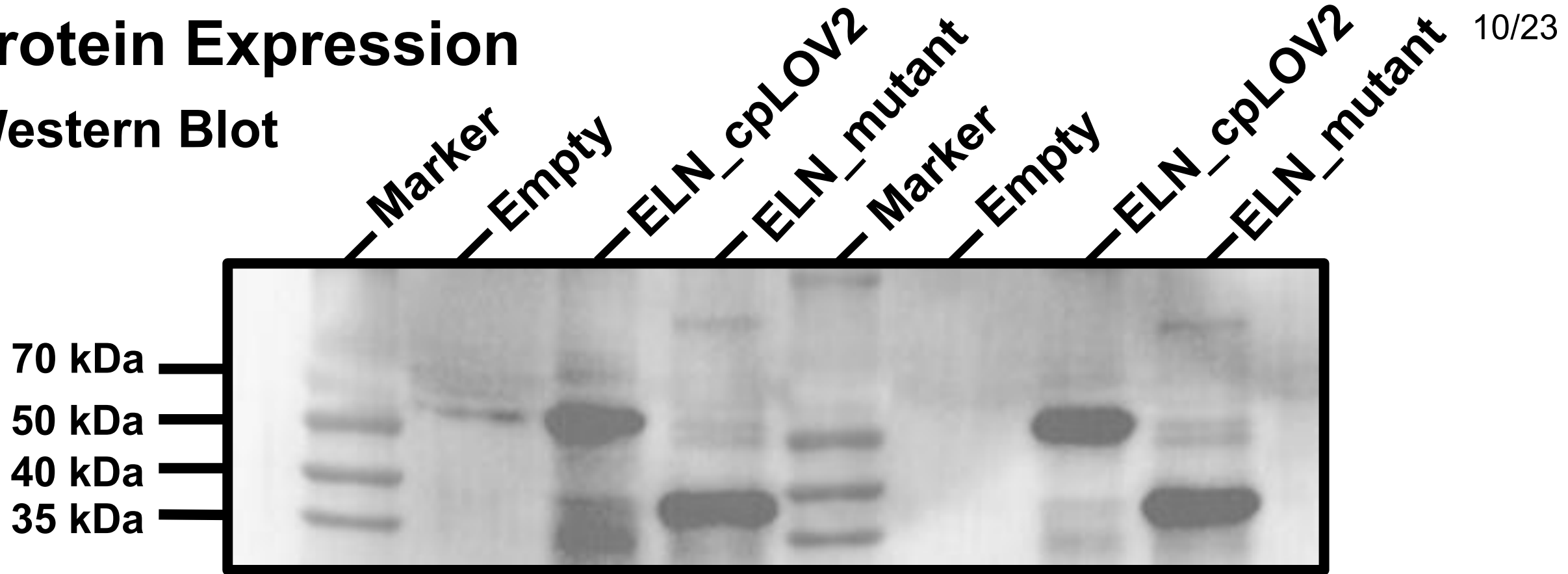
Sanger Sequencing

- Align mutants to the original ELN_cpLOV2 reference
- Complete deletion of cpLOV2 domain in the mutant and successful plasmid construction



Protein Expression

Western Blot

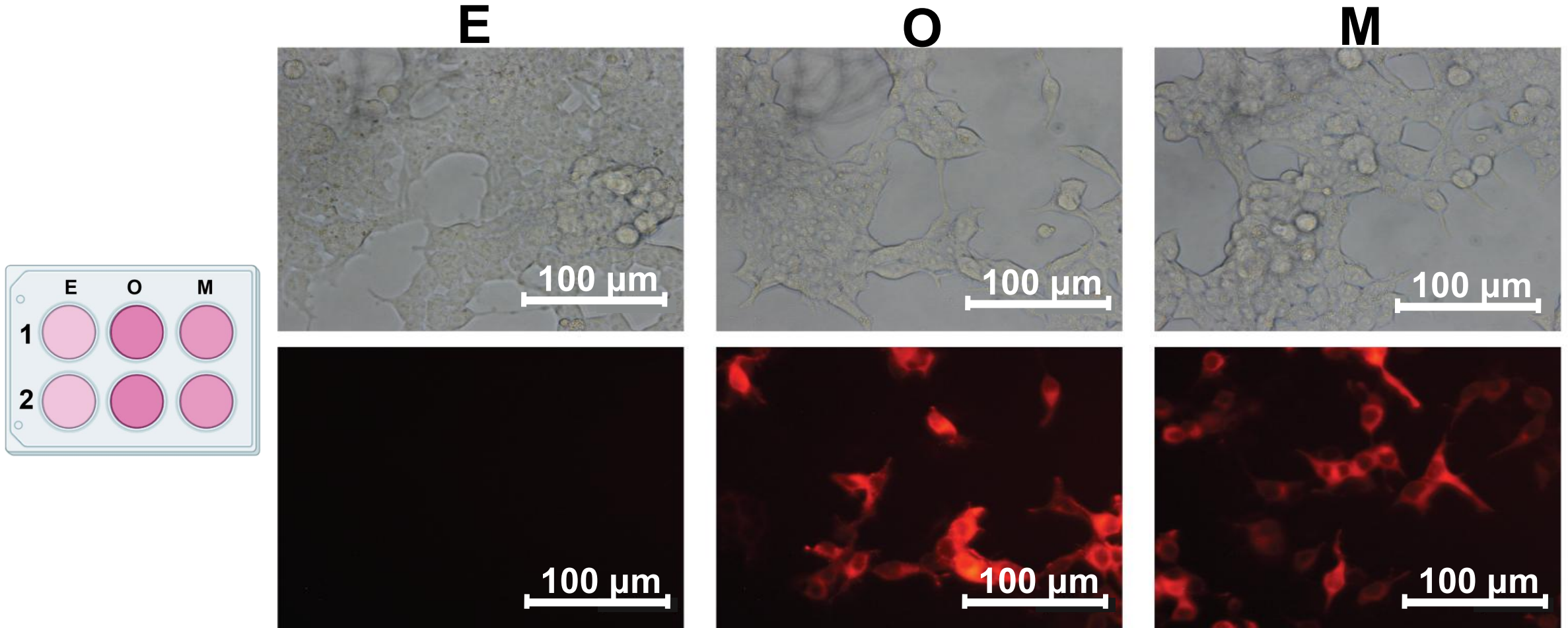


- Utilize anti-FLAG antibody as 1st Antibody
- ELN_cpLOV2 proteins and ELN_mutant proteins expressed in the full-length with high concentrations in cells



Protein Expression 40X Nikon Fluorescence Microscope ^{11/23}

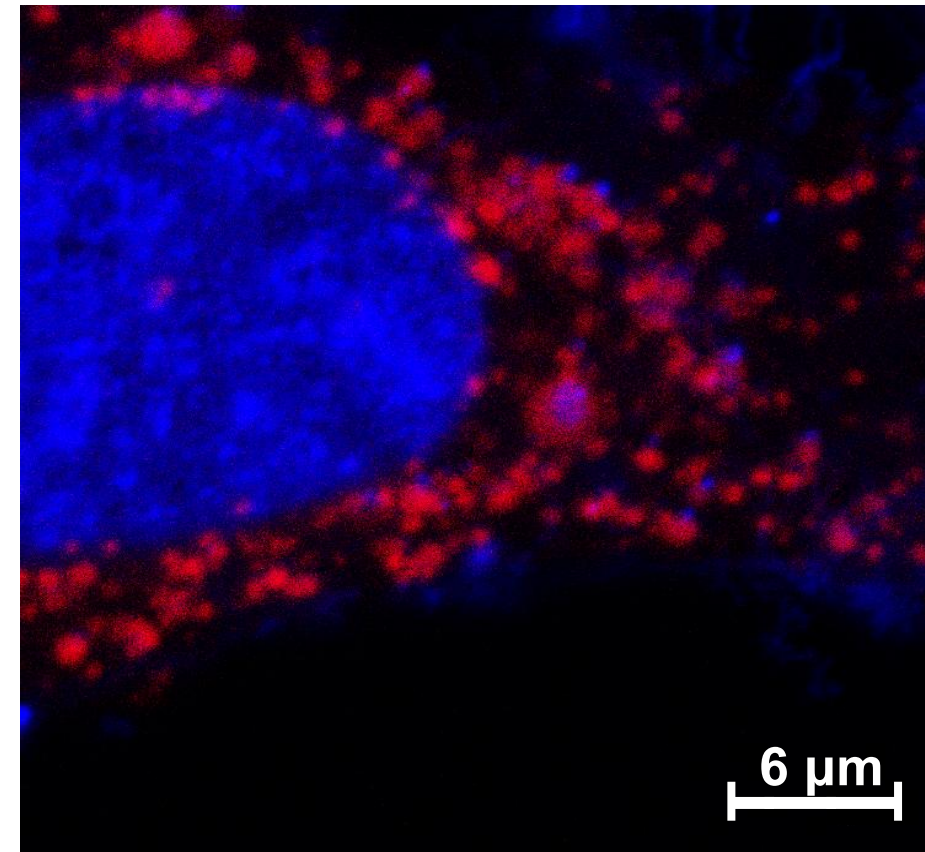
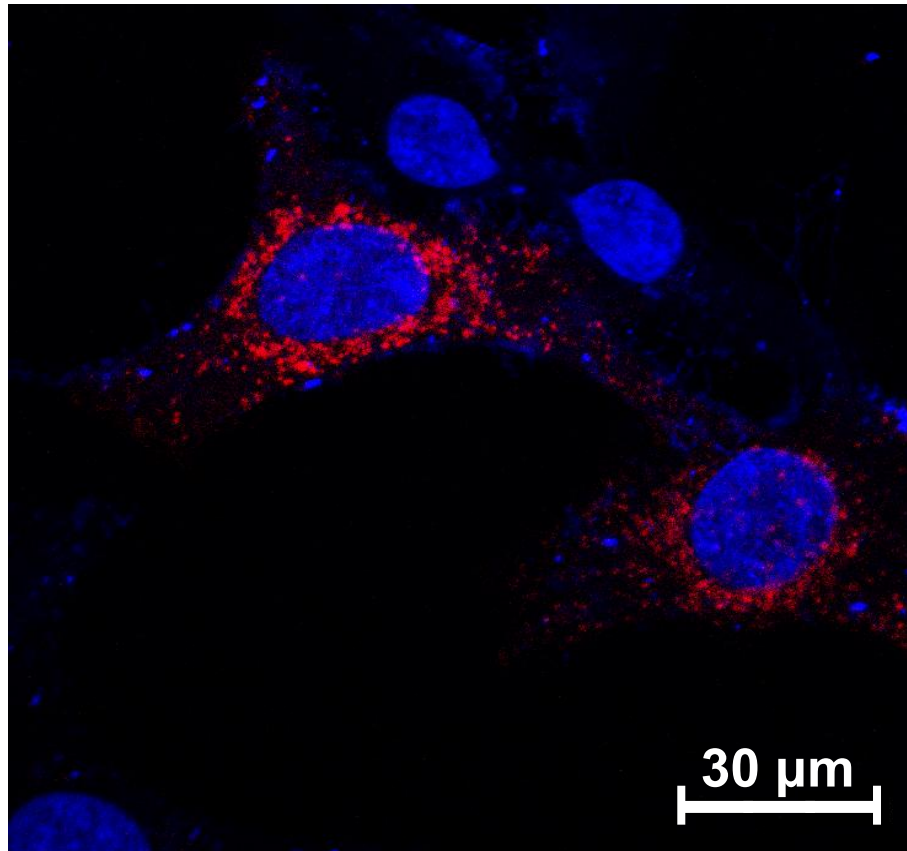
- Recombinant proteins for both origins and mutants were highly expressed in living cells
- Mutant protein groups contained a higher proportion of round shape cells



Protein Distribution 63X Oil Lens Confocal Microscope

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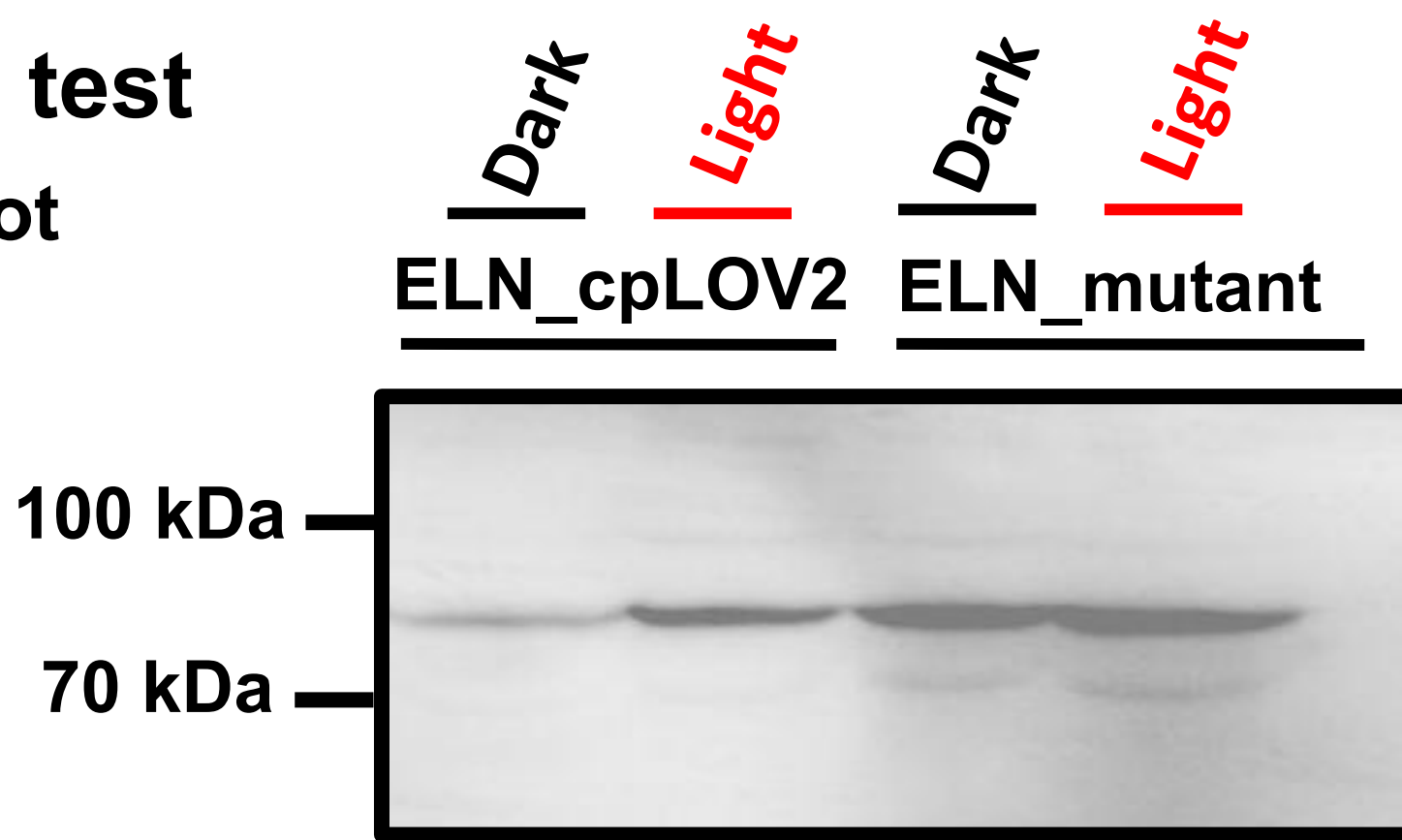
- ELN_cpLOV2 proteins localized around the cell nucleus (stained in blue)
- Proteins did not distribute continuously and gathered to make puncta



ER stress test

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Western Blot



- Utilize anti-GRP78 antibody as 1st Antibody
- GRP78 protein concentrations were higher in light ELN_cpLOV2 group's than dark group
- ELN_mutants induced the high GRP78 concentrations

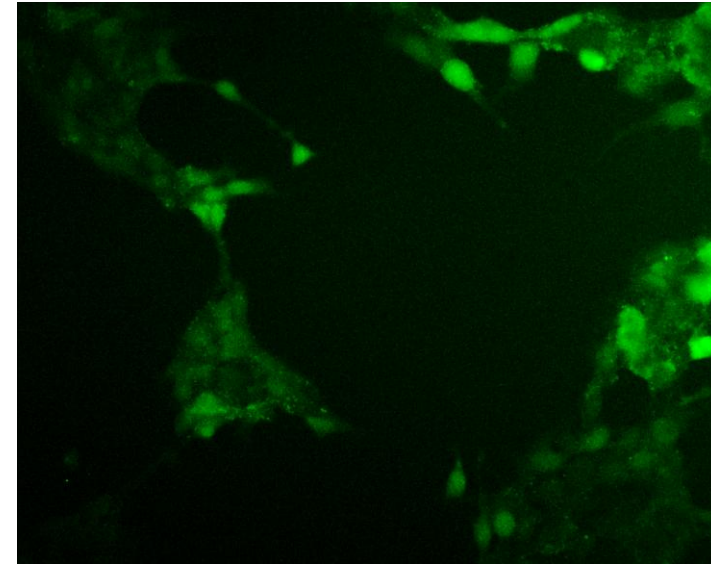
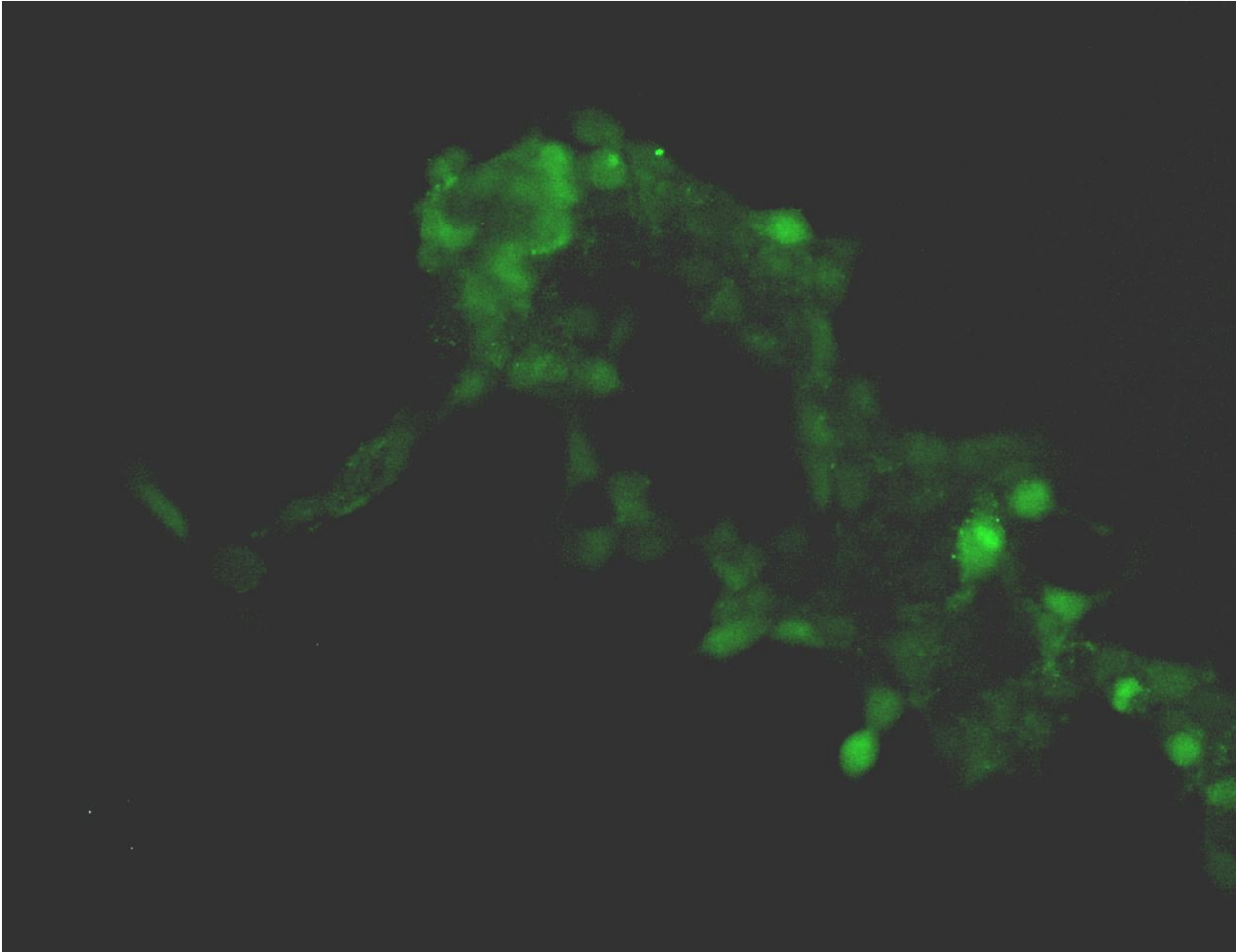


Ca²⁺ Imaging

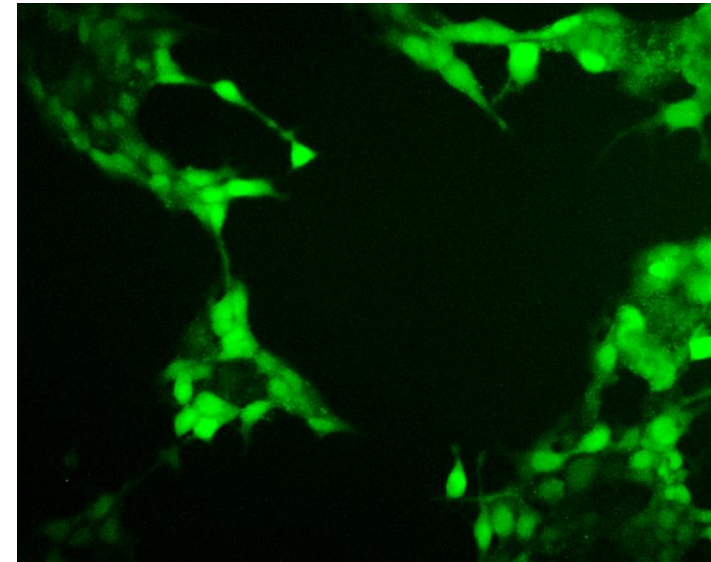
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Calbachol induced Ca²⁺ release from ER

Representative sample (2E sample)



**Before
calbachol
stimulation**

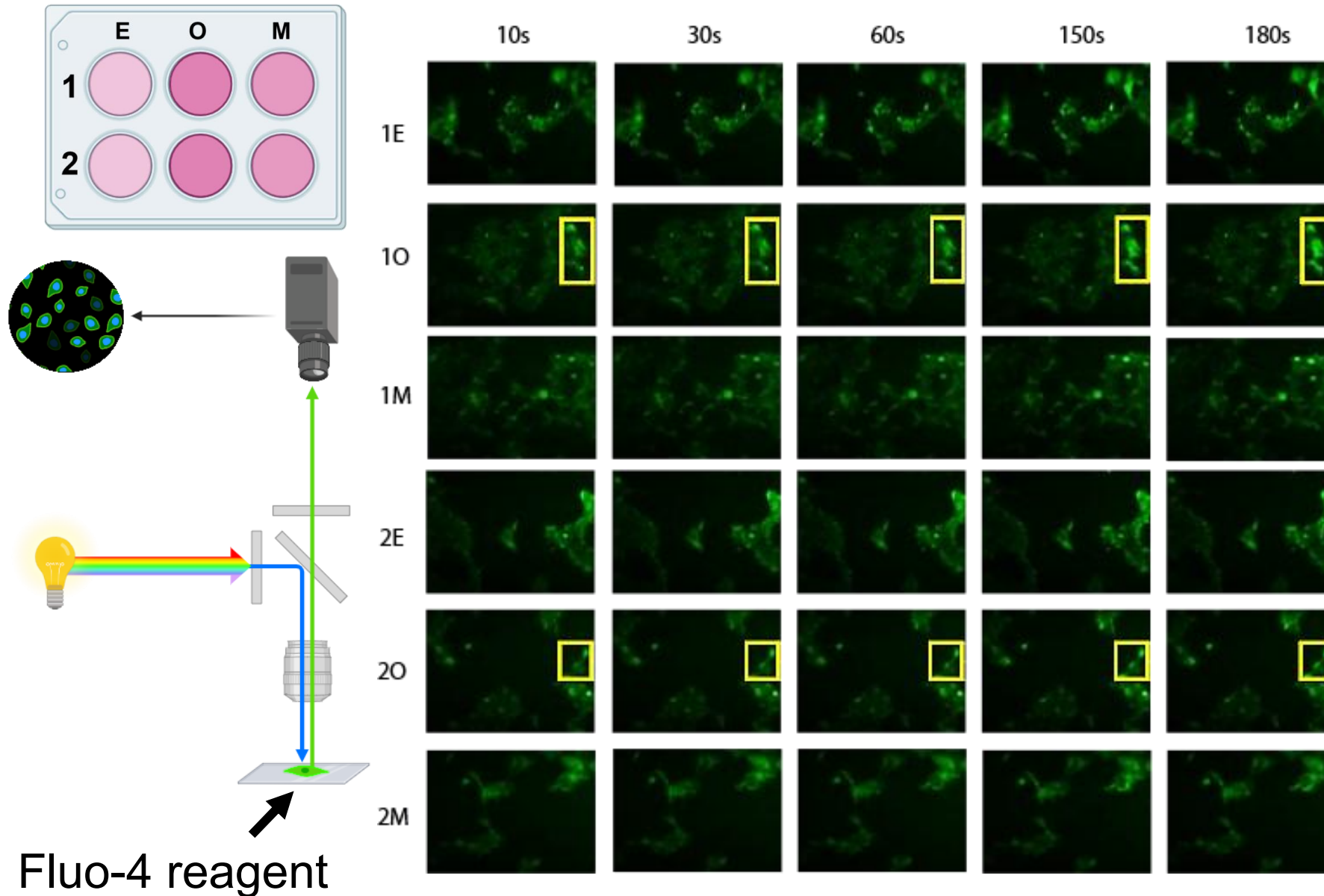


**After
calbachol
stimulation**

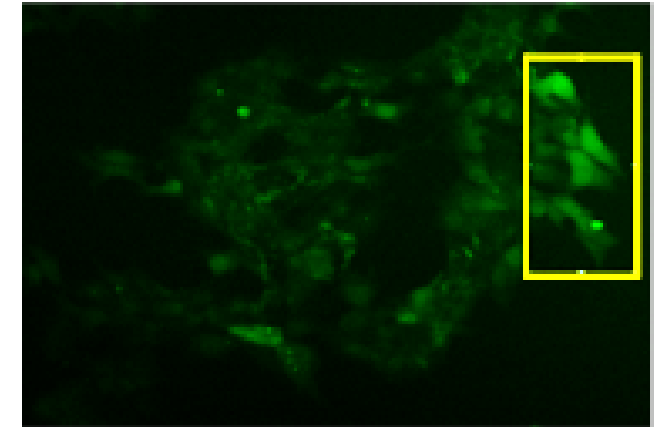


Ca²⁺ Imaging

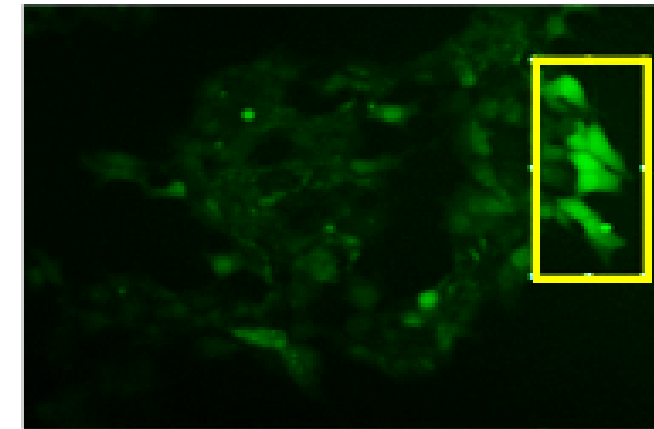
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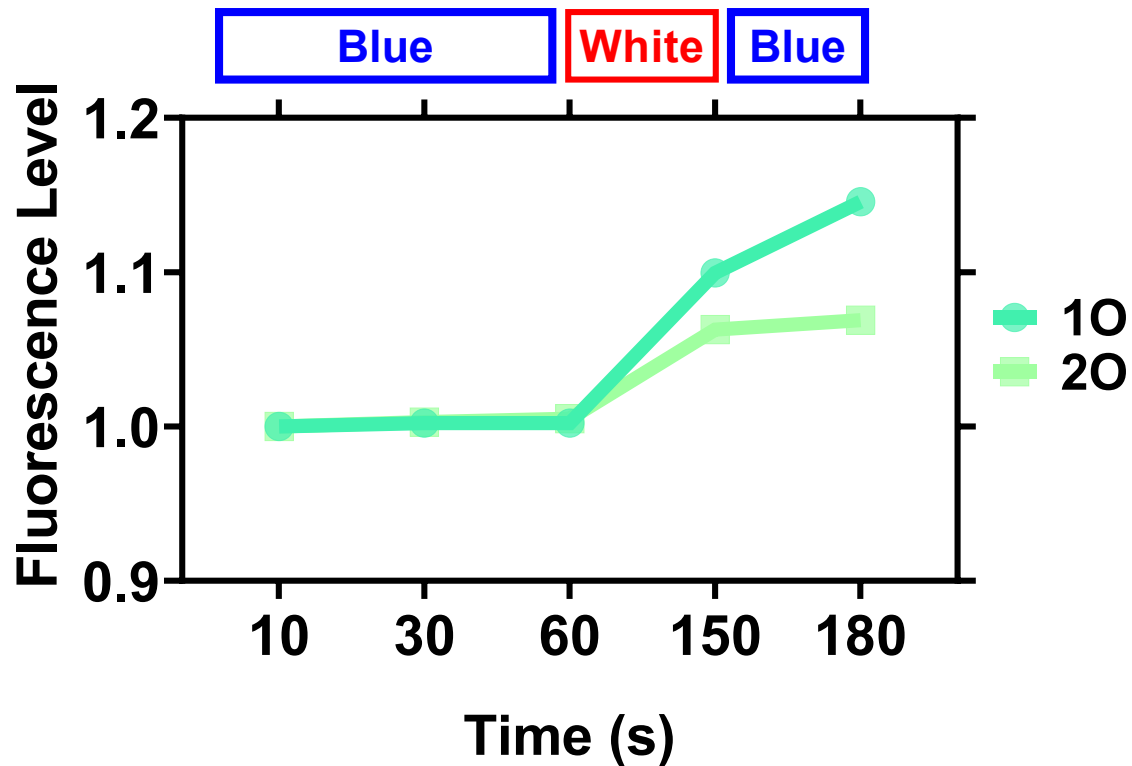


Before light stimulation



After light stimulation





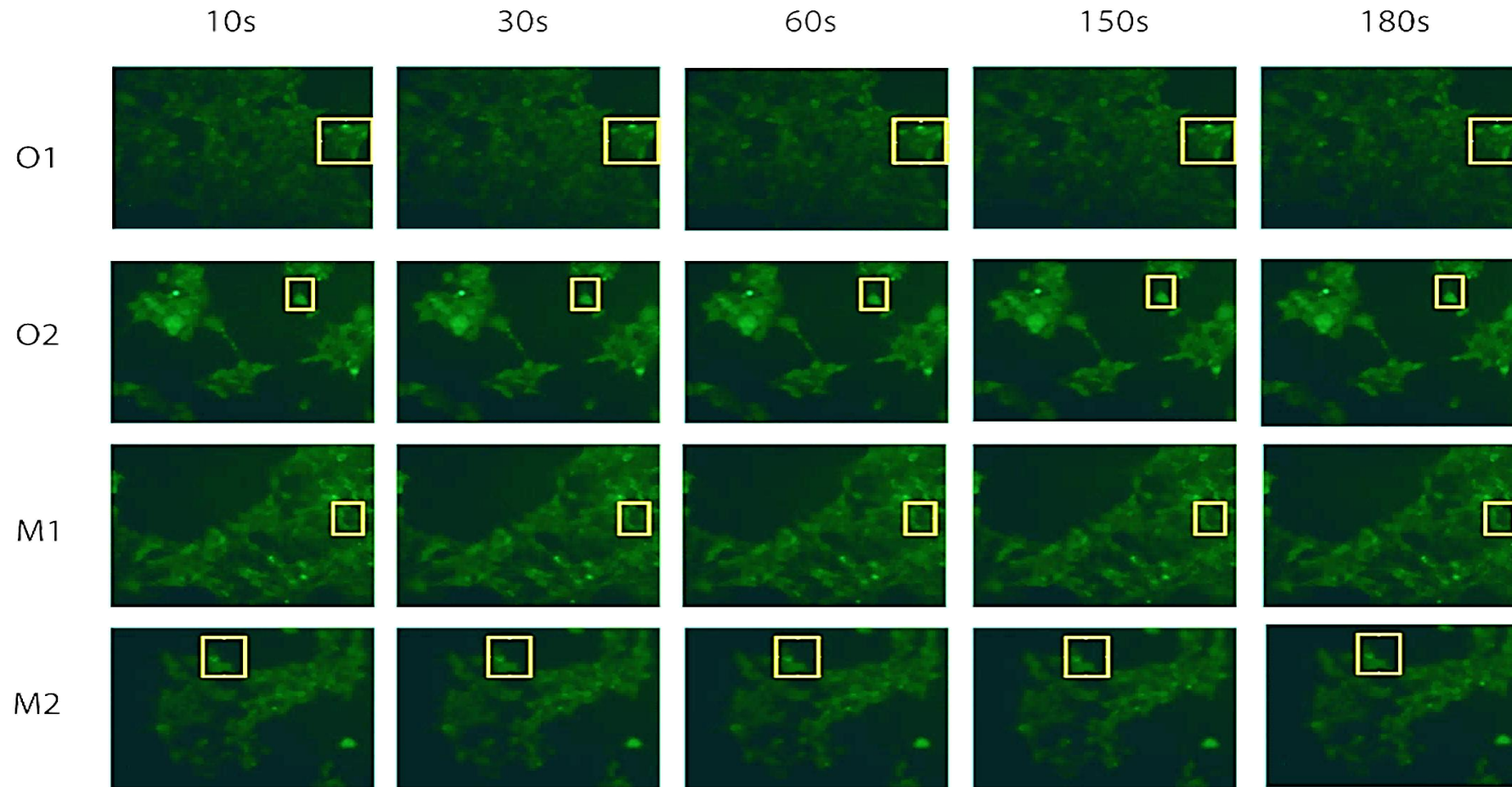
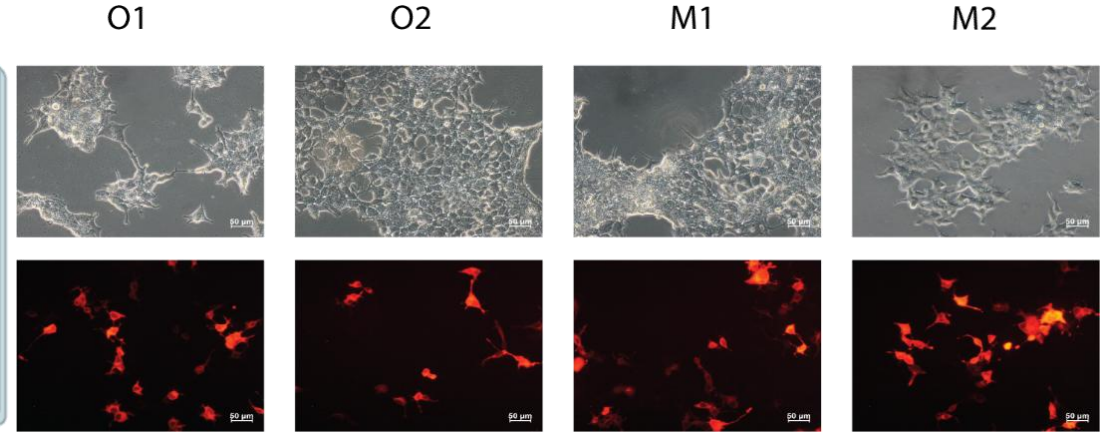
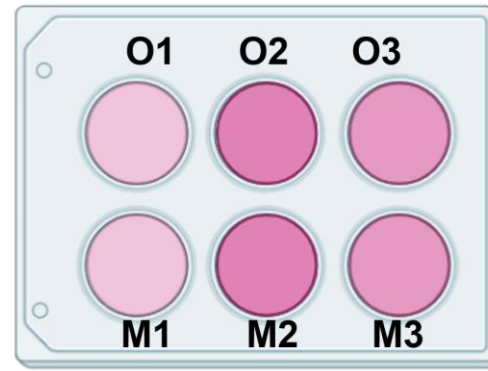
- The relative fluorescent level was measured based on the mean gray value of ROI
- 10s ROI gray value as the initial value and calculated other relative values
- Regions of interests (ROI) displayed sharp fluorescence increase following light stimulation

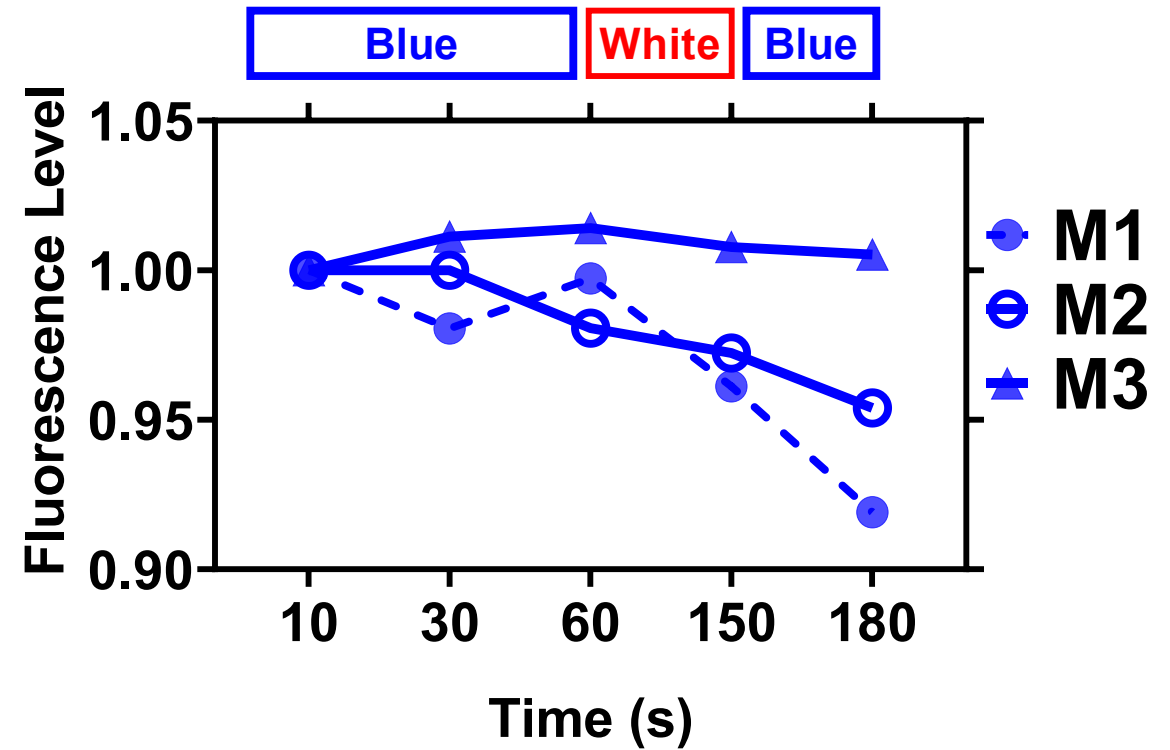
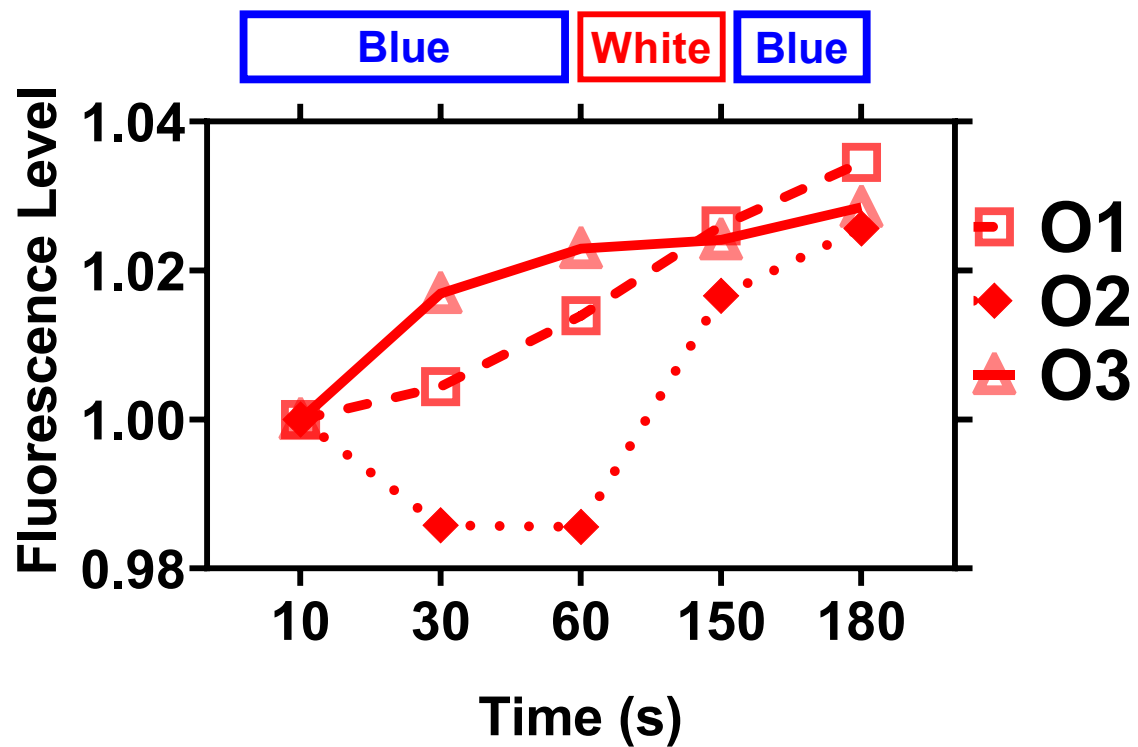


Ca²⁺ Imaging

Re-verification test

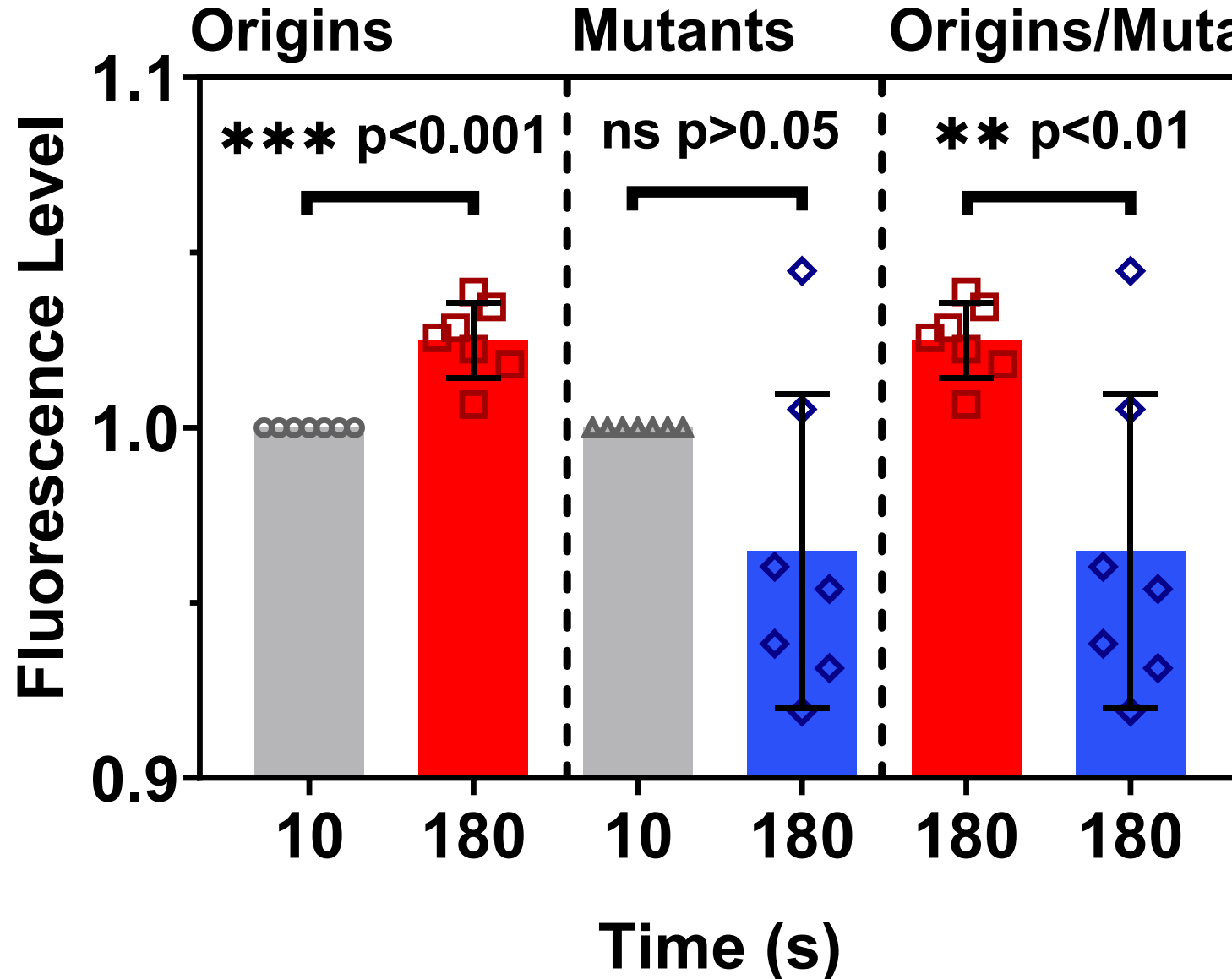
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- Fluorescence intensity of original group samples showed an upward trend under the light stimulation
- Mutant group revealed a conceivable downward trend



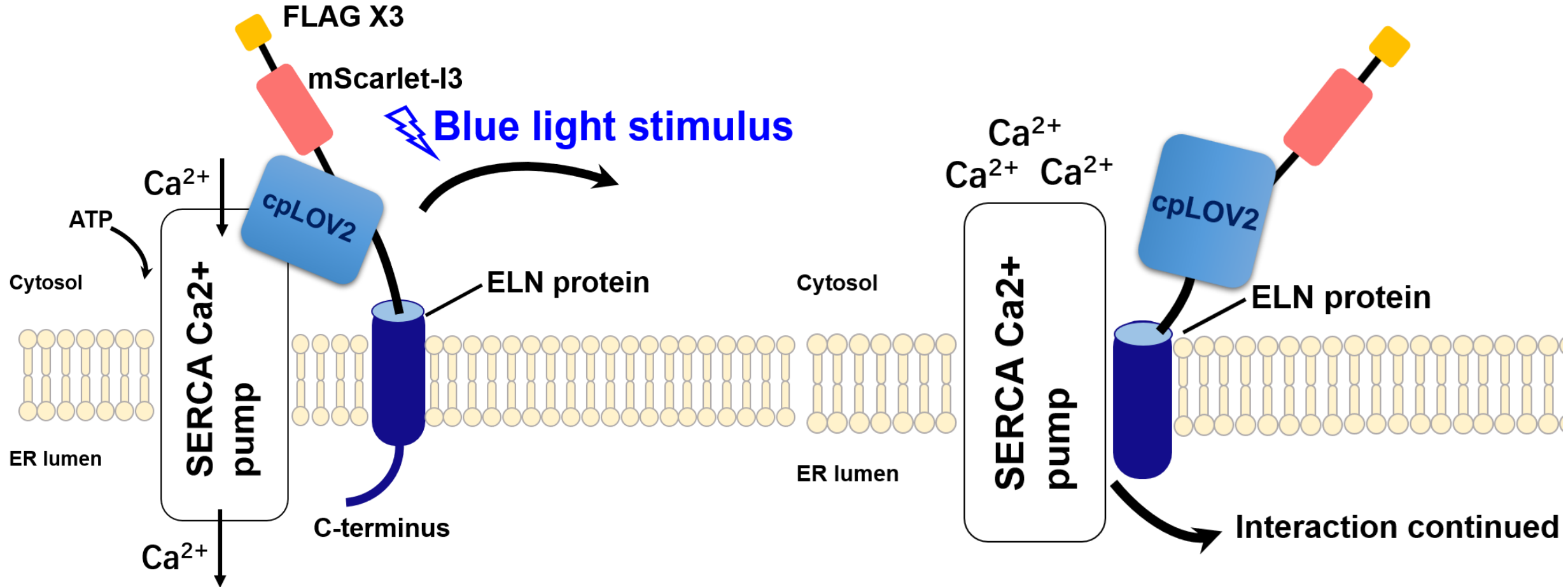


- ELN-cpLOV2 protein might block the SERCA Ca²⁺ pump under light stimulus
- Ca²⁺ transportation into ER lumen was obstructed
- Accumulation of Ca²⁺ in the cell regions



Proposed Model **Support the hypothesis**

20/23



Conclusion

- **Developed a novel optogenetic tool ELN_cpLOV2 to regulate the Ca^{2+} pump SERCA on human endoplasmic reticulum**
- **Verified the feasibility of the photosensitive tool by Ca^{2+} Imaging**
- **Contributions to the cellular ER stress problem and treatment for correlated diseases**



Acknowledgements

22/23

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THANK YOU!

